

Civil Engineering Analysis II

Project-2 SOW

CIVE-202-150

[Project Overview](#)

The Nebraska Department of Transportation (NDOT) has requested a transfer of their current concrete mix design model from an excel formatted sheet to a structured python system. This project aims to automate the calculations used in the excel file, and to improve its reliability. The program will allow the user to input the necessary material properties and ratios, as well as the parameters in a well guided manner to generate the correct weight summary for one cubic yard. The purpose of the project is to build onto previous workflows by implementing personal functions, prompts to guide the user, as well as testing of various scenarios to verify consistency and accuracy.

[Technical Objectives](#)

The primary objective for his project is to fully translate the excel functions and logic into a comprehensible python code. The given calculations that were used in the spreadsheet will be translated and organized into defined python functions. For each given computation (Water weight, Cement content volumes, etc.), each will be modified to improve understanding and consistency. The program will be designed to prompt the user for an input on each necessary input, such as material weights, and specific gravities. The inputs will be collected in a given order similar to the excel sheet to hold consistency. After all inputs are given and results are calculated, the program will produce an organized weight chart for one cubic yard of concrete. The values will be labeled and displayed in a professional table that can be easily read and interpreted.

[Scenario Development and Validation](#)

To validate the reliability and consistency of the model, four scenarios with varying values will be researched and prepped for use. These scenarios may be based on NDOT given information and possibly other established engineering resources. Upon finding the other scenarios, the values will be inputted into the Python model and will display the same information as the original scenario. The results of the outputs will be analyzed for accuracy and consistency. This ensures that this code can be used for numerous situations and can still be reliably used to find the results.

[Deliverables](#)

The deliverable for this project will ultimately include:

- A functional Python program that represents NDOT's Mix Design excel sheet
- Defined functions for calculation of each major mix component
- An organized user input workflow
- A formatted output chart displaying weights and units for one cubic yard of concrete
- Tested results from four different concrete mix scenarios

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